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SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	7 July, 2026
Team ID	SWTID-2026-8420
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality of the Credit Card Approval Prediction project. The project is developed using Machine Learning and Flask to automate the credit card approval process. The code is modular, well-documented, follows Python coding standards, and satisfies the functional requirements of the system.

Solution Summary

Field	**Details**
Repository Link / URL	https://github.com/jagannagari/AI-ML-SmartbridgeProject-Credit-Card-Approval-Prediction-
Programming Language(s)	Python 3.11, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Pickle
Key Features Implemented	• Data preprocessing and cleaning • Feature encoding and scaling • Model training using Logistic Regression, Decision Tree, Random Forest, and XGBoost • Best model saved using Pickle • Flask-based web application for real-time prediction • User-friendly input form for applicant details • Instant credit card approval/rejection prediction
Pending / Incomplete Features	• User login authentication • Database integration for storing prediction history • Email/SMS notification system • Cloud deployment (optional enhancement)
Setup / Run Instructions	1. Install Python libraries using pip install -r requirements.txt. 2. Place model.pkl in the project folder. 3. Run python app.py. 4. Open http://127.0.0.1:5000 in a web browser. 5. Enter applicant details and click Predict to view the result.

Code Quality Checklist

S.No	Criteria	Status(yes/no)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The Credit Card Approval Prediction project follows good software engineering practices by maintaining a modular architecture, reusable code components, and clear documentation. Machine learning algorithms are trained and evaluated to select the bestperforming model, which is integrated into a Flask web application for real-time prediction. The application is scalable and can be enhanced further by adding database support, authentication, cloud deployment, and prediction history management